Disaster Management

Course Code: 22CE409

DISASTER PREPAREDNESS, PLANNING AND MANAGEMENT

(Open Elective-I: Common to all branches except CE)

Instruction	:	3 Periods / week	Continuous Internal Evaluation	:	40 Marks
Tutorial	:	0 Period / week	Semester End Examination	:	60 Marks
Credits	:	3	Semester End Exam Duration	:	3 Hours

Prerequisites:

Course Objectives: The objectives of the course are

1. To Understand basic concepts in Disaster Management.
2. To Understand Definitions and Terminologies used in Disaster Management.
3. To Understand Types and Categories of Disasters.
4. To Understand the Challenges posed by Disasters.
5. To understand Impacts of Disasters Key Skills.

Unit-I

Introduction- Concepts and definitions: disaster, hazard, vulnerability, resilience, risks severity, frequency and details, capacity, impact, prevention, mitigation, disaster phenomena, events-global National & Regional.

Unit-II

Disasters- Disasters classification; natural disasters (floods, draught, cyclones, volcanoes, earthquakes, tsunami, landslides, coastal erosion, soil erosion, forest fires etc.); manmade disasters(industrial pollution, artificial flooding in urban areas, nuclear radiation, chemical spills, transportation accidents, terrorist strikes, etc.); hazard and vulnerability profile of India, Caved 2019 in India, mountain and coastal areas, ecological fragility, coping with disaster-strategies, safety norms &survival kits.

Unit-III

Disaster Impacts- Disaster impacts (environmental, physical, social, ecological, economic, political, etc.); health, psycho-social issues; demographic aspects (gender, age, special needs); hazard locations; global and national disaster trends; climate change and urban disasters, capacity building – concepts, assessment –structural & non-structural measures, legislative support.

Unit-IV

Disaster Risk Reduction (DRR) - Disaster management cycle – its phases; prevention, mitigation, preparedness, relief and recovery; structural and non-structural measures; risk analysis, vulnerability and capacity assessment; early warning systems, Post disaster environmental response (water, sanitation, food safety, waste management, disease control, security, communications); Roles and responsibilities of government, community, local institutions, NGOs and other stakeholders; Policies and legislation for disaster risk reduction, DRR programmes in India and the activities of National Disaster Management Authority.

Unit-V

Disasters, Environment and Development- Factors affecting vulnerability such as impact of developmental projects and environmental modifications (including of dams, land use changes, urbanization etc.), sustainable and environmental friendly recovery; reconstruction and development methods.

Course Outcomes: The student will develop competencies in

CO 1 : The application of Disaster Concepts to Management.
CO 2 : Analyzing Relationship between Development and Disasters.
CO 3 : Ability to understand Categories of Disasters.
CO 4 : Realization of the responsibilities to society.
CO 5 : Ability to understand towards sustainable approach.

Textbooks:

1. Pradeep Sahni, 2004, Disaster Risk Reduction in South Asia, Prentice Hall.
2. Singh B.K., 2008, Handbook of Disaster Management: Techniques & Guidelines, Rajat Publication.

References:

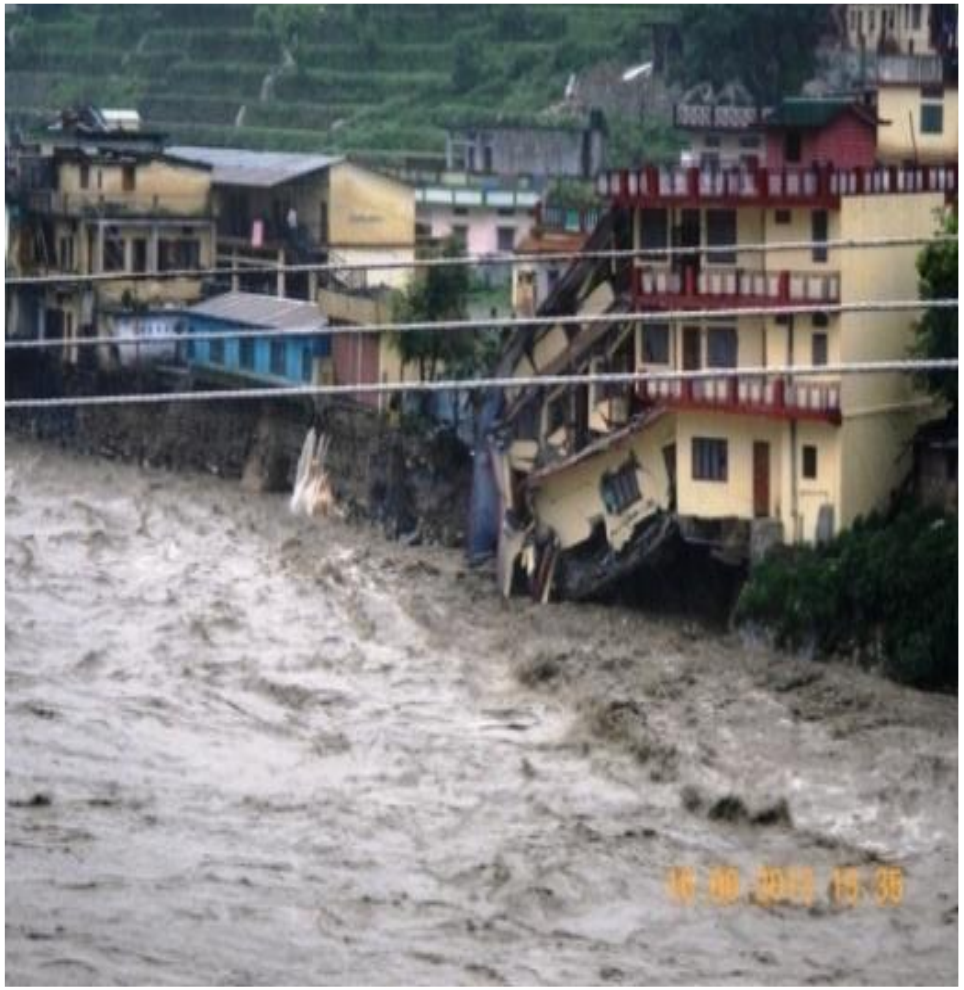
1. Disaster Medical Systems Guidelines. Emergency Medical Services Authority, State of California, EMSA no.214, June 2003.
2. Inter-Agency Standing Committee (IASC) (Feb. 2007). IASC Guidelines on Mental Health and Psychosocial Support in Emergency Settings. Geneva: I ASC.
3. Disaster Management, Ghosh G.K., APH Publishing Corporation, 2011.

- As per the Oxford dictionary a disaster is “a sudden accident or a natural catastrophe that causes great damage or loss of life”

What is a Disaster ?

- The term disaster owes its origin to the French word “Desastre” which is a combination of two words ‘des’ meaning bad and ‘aster’ meaning star.

Thus the term refers to ‘Bad or Evil star’.



What is a Disaster ?

- A disaster can be defined as

“A serious disruption in the functioning of the community or a society causing wide spread material, economic, social or environmental losses which exceed the ability of the affected society to cope using its own resources”.

DISASTER

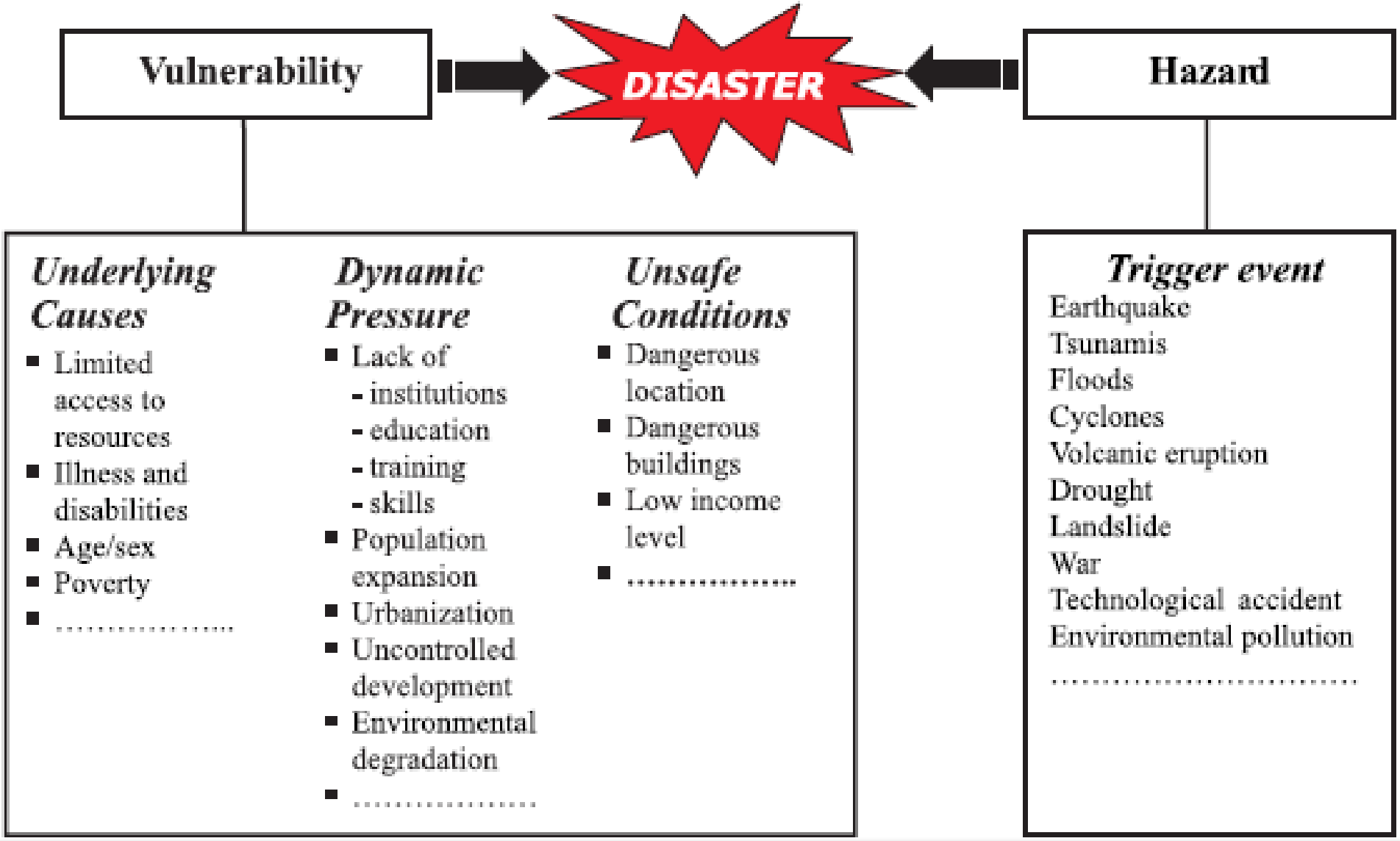
- A Disaster is an event or series of events, which gives rise to casualties and damage or loss of properties, infrastructure, environment, essential services or means of livelihood on such a scale which is beyond the normal capacity of the affected community to cope with.

- The United Nations defines disaster as “the occurrence of sudden or major misfortune which disrupts the basic fabric and normal functioning of the society or community”.

DISASTER

- A disaster is a result from the combination of hazard, vulnerability and insufficient capacity or measures to reduce the potential **chances** of risk.

- A disaster happens when a hazard impacts on the vulnerable population and causes damage, casualties and disruption.


Vulnerability
DISASTER
Hazard
Underlying Causes
Limited access to resources
Illness and disabilities
Age/sex
Poverty
……..
Dynamic Pressure
Lack of
- institutions
- education
- training
- skills
Population expansion
Urbanization
Uncontrolled development
Environmental degradation
……
Unsafe Conditions
Dangerous location
Dangerous buildings
Low income level
……..
Trigger event
Earthquake
Tsunamis
Floods
Cyclones
Volcanic eruption
Drought
Landslide
War
Technological accident
Environmental pollution
……

What is a Hazard ?

- Hazard may be defined as “a dangerous condition or event, that threat or have the potential for causing injury to life or damage to property or the environment.”

- The word ‘hazard’ owes its origin to the word ‘hasard’ in old French and ‘az-zahr’ in Arabic meaning ‘chance’ or ‘luck’.

According to NIDM

- **Hazard:** A dangerous phenomenon, substance, human activity or condition that may cause loss of life, injury or other health impacts, property damage, loss of livelihoods and services, social and economic disruption, or environmental damage.

How is it classified?

- Hazards can be grouped into two broad categories namely natural and manmade.

How is it classified?

- **1. Natural hazards** are hazards which are caused because of natural phenomena (hazards with meteorological, geological or even biological origin).

- Examples of natural hazards are cyclones, tsunamis, earthquake and volcanic eruption which are exclusively of natural origin.

- Landslides, floods, drought, fires are socio-natural hazards since their causes are both natural and man made. For example, flooding may be caused because of heavy rains, landslide or blocking of drains with human waste.

How is it classified?

- **2. Manmade hazards** are hazards which are due to human negligence.

- Manmade hazards are associated with industries or energy generation facilities and include explosions, leakage of toxic waste, pollution, dam failure, wars or civil strife etc.

Various types of hazards

TYPES	HAZARDS
Geological Hazards	1. Earthquake 2. Tsunami 3. Volcanic eruption 4. Landslide 5. Dam burst 6. Mine Fire
Water & Climatic Hazards	1. Tropical Cyclone 2. Tornado and Hurricane 3. Floods 4. Drought 5. Hailstorm 6. Cloudburst 7. Landslide 8. Heat & Cold wave 9. Snow Avalanche 10.Sea erosion

Various types of hazards

TYPES	HAZARDS
Environmental Hazards	1. Environmental pollutions 2. Deforestation 3. Desertification 4. Pest Infection
Biological	1. Human / Animal Epidemics 2. Pest attacks 3. Food poisoning 4. Weapons of Mass Destruction
Chemical, Industrial and Nuclear Accidents	1.Chemical disasters 2. Industrial disasters 3. Oil spills/Fires 4. Nuclear

Various types of hazards

TYPES	HAZARDS
Accident related	1. Boat / Road / Train accidents / air crash/ Rural or Urban fires/ Bomb /serial bomb disasters Blasts 2. Forest fires 3. Building collapse 4. Electric Accidents 5. Festival related 6. Mine flooding

What is vulnerability ?

- Vulnerability may be defined as “The extent to which a community, structure, services or geographic area is likely to be damaged or disrupted by the impact of particular hazard, on account of their nature, construction and proximity to hazardous terrains or a disaster-prone area.”

What is vulnerability ? CLASSIFICATION

- Vulnerabilities can be categorized into physical and socio-economic vulnerability.
- Physical Vulnerability: It includes notions of **who** and **what** may be damaged or destroyed by natural hazard such as earthquakes or floods.
- Socio-economic Vulnerability: The degree to which a population is affected by a hazard will not merely lie in the physical components of vulnerability but also on the socio-economic conditions.
- **Vulnerability:** The characteristics and circumstances of a community, system or asset that make it susceptible to the damaging effects of a hazard.

What is capacity ?

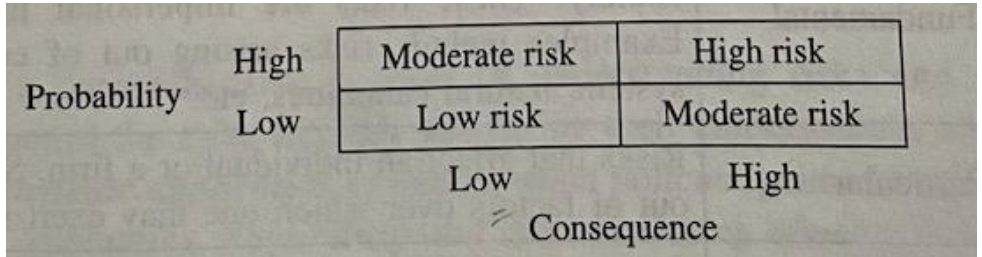
- Capacity can be defined as “resources, means and strengths which exist in households and communities and which enable them to cope with, withstand, prepare for, prevent, mitigate (lessen)or quickly recover from a disaster”.

- **Physical Capacity:** People whose houses have been destroyed by the cyclone or crops have been destroyed by the flood can salvage(prevent from being ruined) things from their homes and from their farms.

- **Socio-economic Capacity:** In most of the disasters, people suffer their greatest losses in the physical and material realm.(Rich/Poor)

Risk

- It is associated with the degree to which humans cannot cope with a particular event or situation,
- It is the combination of the probability of an event, and the negative consequences arising out of it.
- According to CSA (1997) risk is *'the chance of injury or loss as defined as a measure of the probability and severity of an adverse effect to health, property, the environment, or other things of value'*.
- The intensity of a risk is based on the social context in which it occurs.
- Risk = Probability × Consequences
- (Conscious risk, knowledge of risk is not eliminate risk)



Probability	High	Moderate risk	High risk
	Low	Low risk	Moderate risk
		Low	High
		Consequence	

Figure 14.1 Intensity of risk grid.

Disaster Risk

- The UNISDR (2009) defines *disaster risk as potential disaster losses, in lives, health status, livelihoods, assets and services, which could occur to a particular community or a society over some specified future time period.*
- Disaster risk is created by a combination of exposure, vulnerability and hazard risk
- The effect of hazards on the population varies substantially with the level of coping mechanisms available within that particular community.

$$\text{Disaster risk (R)} = \frac{\text{Vulnerability (V)} \times \text{Hazard (H)}}{\text{Capacity (C)}}$$

Disaster risk management

- *Disaster risk management attempt to lessen, avoid or transfer adverse effects of hazards through strategic management activities, like prevention, mitigation and preparedness.* *(structural and non-structural measures)*
- Disaster risks can be reduced through:

1. Systematic analysis and management of the factors that causes disasters,
2. Reduction in the exposure to hazards,
3. Lessening the vulnerability of people and property,

Impact of Disasters

- Disasters can affect a nation's stability and its standing and image before the international community.
- The population expects a lot from their respective Governments at times of disasters.

EFFECTS ON CURRENT DEVELOPMENT

- Loss of crops
- Loss of livestock (avian flu (2013, 2014))
- Degradation of land (Ayila cyclone, Tsunami)
- Infrastructure loss

    Loss of roads and bridges (Curb the transportation)

    Loss of buildings and facilities (affects developmental work)

    Damage to harbours, wharfs and landing yards

    Loss to airport and aircrafts

EFFECTS ON LONG-TERM DEVELOPMENT

- A disaster could lead to material and economic losses.
- For instance, a disaster stricken *coconut plantation* could require near to a decade to be replaced.
- There is also the danger of long-term unsuitability of land areas.
- There could be the cost of relocating of the communities involved.
- The vital resources include communication facilities, power supplies, roads and transportation, housing, water supplies, medical facility, loss of productivity of soils, etc.

- The planning process could be seriously upset in the event of a disaster striking. Causalities with respect to the planning process include:

1. Disruption of the current plan.
2. Unanticipated delays in the planned process due to unexpected pressure on the available resources.
3. Impossibility of predicting and forecasting development progress and its likely status at the end plan period.
4. Disasters would also place impediments (delays) in the formulation and contents of the subsequent national development plans.

Mitigation

- Mitigation is the set of actions taken well in advance of disasters to avoid or reduce disaster-related damage.
- According to the NRC (2006) ‘*hazard mitigation consists of practices that are implemented before impact and provide passive protection at the time impact occurs'.*
- At the community or local government level mitigation measures include appropriate land-use practices, coastal zone management practices, building codes, and a host of other long-term loss reduction techniques.
- It could also include preparedness for an event as well as long-term risk reduction measures.

- Asian Development Bank provides a crisp definition, mitigation as *'Measures aimed at reducing the impact of a natural or man-made disaster on a nation or community'*.
- Mitigation decisions consider aspects like costs, politics, past experiences and a number of other related issues.

MITIGATION MEASURES

- Strengthening buildings to make them resistant to cyclones, floods, earthquakes, and other disasters.
- All future developmental structures and buildings to be built in such a manner that hazard resistance are incorporated into it.
- Planting only those types of crops that are less prone to disasters that are likely to strike a particular area.
- Altering cropping patterns and cycles in such a manner that it is possible to harvest the crops prior to the onset of disasters prone seasons like flood or cyclone.
- Adopting land-use patterns and controls such that developmental activities are not carried out in high-risk areas.
- Adoption of economic diversification so that losses in one sector are offset by increased production or output in certain other sectors.
- Education and training of the members of the society and officials dealing with disaster mitigation about the need and modalities of mitigation activities.

Mitigation Strategies

- Maintaining and enhancing environmental quality
- Maintaining and enhancing the quality of life.
- Fostering local resiliency and responsibility for disasters.
- Recognising that sustainable and vital local economies are essential for mitigation.
- Identifying and ensuring both intra- and intergenerational equity.
- Adopting a *consensus-building approach*, involving all levels, starting at the local level.

Requirements for Effective Mitigation

1. A clear and comprehensive national disaster policy and mitigation measures that addresses all related aspects of disaster management. It should ensure that mitigation is given proper consideration and priority at all concerned levels.
2. Proper assessment and monitoring of hazards and its vulnerabilities. This will help in identifying and defining the need for appropriate mitigation measures.
3. Proper support for various forms of traditional mitigation measures, where ever it can be put to best use.
4. Accurate analysis of all reasonable and possible mitigation projects.
5. Readiness and willingness of the respective governments to institute and carry through appropriate mitigation programs.

Disaster Resilience

- Resilience is the continued ability of a person, group, or system to adapt to stress', so that it may continue to function, or quickly recover its ability to function, during and after stress.
- Disaster resilience is the ability of those who are affected by disasters to organise themselves; adapt and recover from hazards, shocks or stresses.
- Learn from past disasters; and reduce their risks in future.
- Resilience can be at any level-local, regional, national, or international.
- The OECD (2013) defined it as: *"the ability of individuals, communities and states and their institutions to absorb and recover from shocks, whilst positively adapting and transforming their structures and means for living in the face of long term changes and uncertainty".*

Disaster Resilience Framework

- Context
- Disturbance
- Capacity to respond (Sensitivity, Adaptive capacity)
- Reaction (Bounce back better, Bounce back, Recover but worse than before)

Disaster resilience is a process since it involves series of aspects that support the capacity of individuals, communities and states to adapt through assets and resources relevant to their particular contexts.

Prevention

<u>The Need to Consider Disaster Prevention:</u>

Disaster managers to consider prevention in the context of the following aspects:

- National development planning;
- National disaster management policy;
- Disaster legislation;
- Counter-disaster planning; and
- Special disaster-related circumstances in which particular national assets may be at risk.

Positive Approaches Toward Prevention

- National Policy
- Legislation
- Assessment and Monitoring
- Planning and Organization
- Public Awareness and Education
- International Assistance

Problem Areas in Prevention

- Traditional Outlooks
- Cost
- Other National Priorities
- Political Aspects
- Development Problems
- Balance in Disaster Management
- Public Apathy

Resources Relevant to Prevention

Resources to achieve identification and analysis of disaster threats

- Academic institutions;
- Research establishments;
- Technical authorities;
- Scientific programs; and
- Government departments and agencies which may have major public safety responsibilities (e.g., those dealing with land, sea, and air transport systems);
- Private sector authorities concerned with programs and projects which may generate potential disaster threats (e.g., companies producing hazardous chemicals);
- Industrial safety organizations; and
- International assistance agencies in connection with development projects of various kinds.

- Resources to assess the potential need for and possibilities of implementing programs of disaster prevention
- Government departments, organizations, and agencies;
- National planning authorities;
- Disaster management authorities; and
- Co-opted specialist institutions.

- Resources to implement disaster prevention programs and measures

➢Government departments (especially those possessing technical

➢Capability and expertise);

➢Private sector companies and contractors;

➢NGO's;

➢Military forces;

➢ International disaster assistance agencies; and

➢Disaster management authorities.

- Resources to support disaster prevention activity

✓ NGOs;

✓Media;

✓General public; and

✓Educational authorities

Frequency in Disaster Management:

- In **disaster management**, **frequency** refers to the **rate or likelihood** at which a disaster or hazardous event occurs within a specific period of time, typically in terms of years, decades, or centuries. It helps in understanding how often a certain type of disaster (e.g., floods, earthquakes, hurricanes) is expected to happen in a given area, based on historical data, patterns, and probabilities.
- Disasters can be categorized based on their frequency:

1. **Frequent Disasters**: Events that occur regularly or annually, such as floods, tropical storms, or wildfires.
2. **Infrequent Disasters**: Events that happen less frequently but have higher severity, such as earthquakes, tsunamis, or large volcanic eruptions.
3. **Rare Disasters**: Extremely low-probability events with catastrophic potential, such as asteroid impacts or supervolcanic eruptions.

- Understanding frequency helps disaster management agencies in **risk assessment, preparedness, and resource allocation**, as it informs them about the likelihood of different hazards, guiding efforts to reduce vulnerabilities and plan for potential events.

Disaster phenomena

- **Disaster phenomena** refer to the **natural or man-made events or processes** that cause or have the potential to cause significant damage, disruption, or loss to life, property, infrastructure, and the environment. These phenomena trigger disasters, which can result in widespread humanitarian, economic, and social impacts.

- Disaster phenomena can be broadly categorized into **natural disasters** (those caused by natural processes) and **man-made or technological disasters** (those caused by human activities or failures).

Categories of Disaster Phenomena:

- **Natural Disasters**: These are events that occur due to natural processes of the Earth, often unpredictable and uncontrollable. Common natural disaster phenomena include:
- **Earthquakes**: Sudden shaking of the ground caused by the movement of tectonic plates.
- **Floods**: Overflow of water onto land, often caused by heavy rainfall, storm surges, or the melting of snow and ice.
- **Tornadoes and Hurricanes**: Violently rotating winds and storms that can cause widespread destruction.
- **Volcanic Eruptions**: Explosive eruptions or slow lava flows caused by the movement of molten rock from beneath the Earth's surface.
- **Wildfires**: Uncontrolled fires that spread rapidly, often fueled by dry conditions, vegetation, and wind.
- **Droughts**: Prolonged periods of abnormally low rainfall leading to water shortages.
- **Landslides and Mudslides**: The downward movement of rock, soil, and debris on slopes, often triggered by heavy rainfall or earthquakes.

1. **Man-made or Technological Disasters**: These disasters are caused by human activity, either directly (e.g., industrial accidents) or indirectly (e.g., urbanization and environmental degradation). They include:
    1. **Industrial Accidents**: Chemical spills, explosions, or toxic leaks from factories or other industrial facilities.
    2. **Nuclear Accidents**: Radiation leaks or explosions at nuclear power plants (e.g., Chernobyl, Fukushima).
    3. **Oil Spills**: Discharges of petroleum products into bodies of water, often caused by accidents or negligence.
    4. **Transportation Accidents**: Large-scale accidents involving planes, ships, trains, or trucks that cause significant damage and loss of life.
    5. **Pandemics**: Disease outbreaks that can spread widely across populations, causing a public health crisis (e.g., COVID-19).
    6. **War and Conflict**: Armed conflicts that lead to destruction, displacement, and loss of life.

Global Level

- **Earthquakes and Tsunamis**: The 2004 Indian Ocean tsunami and the 2011 Tōhoku earthquake and tsunami in Japan are examples of large-scale events that affected multiple countries. These events were triggered by massive underwater earthquakes, resulting in loss of life, infrastructure damage, and long-lasting environmental impacts.
- **Hurricanes/Cyclones/Typhoons**: Severe storms like Hurricane Katrina (2005) in the U.S. and Typhoon Haiyan (2013) in the Philippines wreaked havoc with widespread flooding, wind damage, and destruction. These storms are linked to climate change and rising ocean temperatures.
- **Wildfires**: In 2019–2020, wildfires devastated large swathes of Australia, burning millions of acres, destroying ecosystems, and causing air quality deterioration. Similarly, wildfires in California, USA, and other regions around the world have grown in frequency and intensity.
- **Droughts**: Prolonged droughts in parts of Africa, like the East African drought in 2011, and in Australia (such as the 2019–2020 drought), severely impacted agriculture, water resources, and food security.
- **Volcanic Eruptions**: Volcanic events like the eruption of Mount Tambora in 1815 and Mount Pinatubo in 1991 caused significant global cooling and impacted climate patterns.

National Level (India):

- **Floods**: India experiences regular flooding, particularly during the monsoon season. Major floods include the 2017 Kerala floods, 2018 floods in Assam, and the 2020 Bihar floods, which resulted in loss of life, damage to crops, and displacement of millions.
- **Cyclones**: India's eastern and eastern coastal regions are particularly prone to cyclones, with the 1999 Odisha cyclone, 2013 Phailin, and 2020 Amphan being some of the most destructive. These storms bring high winds, storm surges, and intense rainfall, causing widespread damage.
- **Earthquakes**: India is located on several seismic zones. In 2001, the Gujarat earthquake killed tens of thousands of people and caused extensive damage. The 2011 Sikkim earthquake, though less deadly, also had significant impacts in the northeastern region.
- **Heatwaves**: Heatwaves, especially in northern and central India, have become increasingly common, with severe temperature rises (above 45°C). The 2015 heatwave, which claimed over 2,500 lives, highlighted the growing threat of extreme temperatures.

- **Landslides**: Particularly in the Himalayan region and the Western Ghats, landslides are frequent during the monsoon season. These are often triggered by heavy rainfall, as seen in the 2013 Uttarakhand floods, which caused massive landslides and widespread devastation.
- **Wildfires**: Forest fires are common in states like Uttarakhand, Himachal Pradesh, and Maharashtra. These are worsened by climate change, dry conditions, and deforestation.
- **Droughts**: India has faced recurrent droughts, most notably the 2016 drought, which severely affected states like Maharashtra, Karnataka, and Andhra Pradesh. Water shortages, crop failures, and migration are common outcomes.

Regional Level (India)

1. **Coastal Regions** (e.g., Odisha, Andhra Pradesh, West Bengal, Tamil Nadu): Prone to cyclones, storm surges, and coastal flooding. The Bay of Bengal is particularly active during the cyclone season.
2. **Northeastern India** (e.g., Assam, Arunachal Pradesh): Earthquakes, floods, and landslides are common. The region is also vulnerable to flash floods during the monsoon.
3. **Himalayan Region** (e.g., Jammu & Kashmir, Himachal Pradesh, Uttarakhand): Landslides, glacial lake outburst floods (GLOFs), and earthquakes are significant threats. The region is also prone to avalanches during winter.
4. **Western India** (e.g., Gujarat, Rajasthan, Maharashtra): Droughts, heatwaves, and floods are frequent. Gujarat is also vulnerable to earthquakes, as seen in the 2001 Bhuj earthquake.
5. **Southern India** (e.g., Kerala, Tamil Nadu, Karnataka): Floods during the monsoon, cyclones, and occasional droughts affect this region. Kerala also faces landslides due to heavy rainfall.